

Problem Statement: Multi-Agent Agriculture Crop Yield Prediction System

Agriculture is one of the most important sectors for food production and economic growth. However, farmers often face challenges such as unpredictable weather conditions, changing climate patterns, soil fertility variations, pest and disease outbreaks, and inefficient resource management. These factors make it difficult to estimate crop yield accurately and select the most suitable farming practices. As a result, farmers may experience reduced productivity, financial losses, and inefficient use of agricultural resources.

Traditional crop yield prediction methods are often based on historical observations or manual assessments, which may not consider multiple influencing factors simultaneously. Moreover, agricultural information is scattered across various sources, making it difficult for farmers to obtain timely and reliable guidance. There is a need for an intelligent system that can analyze diverse agricultural data, provide accurate crop yield predictions, and support farmers in making informed decisions.

To address these challenges, the proposed **Multi-Agent Agriculture Crop Yield Prediction System** leverages **Agentic AI, IBM Granite Models, LangFlow, and Retrieval-Augmented Generation (RAG)** to create an intelligent decision-support platform. The system employs multiple specialized AI agents that collaboratively analyze soil properties, weather conditions, historical crop data, and agricultural knowledge. Based on this

analysis, the system predicts crop yield, recommends suitable crops, provides weather-based advisories, identifies potential pest and disease risks, and delivers personalized farming recommendations.

The goal of the proposed solution is to improve agricultural productivity, optimize resource utilization, reduce farming risks, and promote sustainable agriculture through accurate, explainable, and data-driven decision-making. This intelligent multi-agent approach helps farmers maximize crop yield and profitability while adapting to changing environmental conditions.